Name: _____ Register No: _____

Experiment 7

Question: Random Sampling and Sampling Distribution.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

def exp_7():
    rng = np.random.default_rng(10)

    # ---- Population: 10,000 customers' monthly spend (right-skewed) ----
    population = pd.DataFrame({
        "CustomerID": np.arange(1, 10001),
        "Spend": rng.exponential(scale=500, size=10000).round(2),
        "City": rng.choice(["Chennai", "Mumbai", "Delhi", "Pune"], 10000,
                           p=[0.4, 0.3, 0.2, 0.1]),
    })
    pop_mean, pop_std = population["Spend"].mean(), population["Spend"].std(ddof=0)
    print(f"Population size : {len(population)}")
    print(f"Population mean : {pop_mean:.2f}")
    print(f"Population std  : {pop_std:.2f}\n")

    n = 100  # sample size

    # ---- Part A: Random sampling methods ----
    srs = population.sample(n=n, random_state=1)                    # simple random
    srs_wr = population.sample(n=n, replace=True, random_state=1)   # with replacement
    k = len(population) // n                                        # systematic
    start = rng.integers(0, k)
    systematic = population.iloc[start::k]
    stratified = population.groupby("City").sample(frac=0.01, random_state=1)  # stratified (1%)

    print("Sampling method comparison (sample mean of Spend)")
    methods = {"Simple random (without replacement)": srs,
               "Simple random (with replacement)": srs_wr,
               "Systematic": systematic,
               "Stratified by City (1% each)": stratified}
    for name, s in methods.items():
        print(f"{name:38s} size={len(s):3d}  mean={s['Spend'].mean():8.2f}")
    print("\nCity proportions - population vs stratified sample")
    print(pd.DataFrame({
        "Population": population["City"].value_counts(normalize=True).round(3),
        "Stratified": stratified["City"].value_counts(normalize=True).round(3)}))

    # ---- Part B: Sampling distribution of the sample mean ----
    num_samples = 1000
    sample_means = np.array([population["Spend"].sample(n, random_state=i).mean()
                             for i in range(num_samples)])
    se_theory = pop_std / np.sqrt(n)
    print(f"\nSampling distribution of the mean ({num_samples} samples, n={n})")
    print(f"Mean of sample means : {sample_means.mean():.2f}  (population mean = {pop_mean:.2f})")
    print(f"Std of sample means  : {sample_means.std(ddof=1):.2f}")
    print(f"Theoretical SE = sigma/sqrt(n) : {se_theory:.2f}")

    # ---- Plots ----
    fig, ax = plt.subplots(1, 2, figsize=(12, 4.5))
    sns.histplot(population["Spend"], bins=40, ax=ax[0], color="steelblue")
    ax[0].axvline(pop_mean, color="red", linestyle="--", label="Mean")
    ax[0].set_title("Population Distribution (skewed)")
    ax[0].legend()
    sns.histplot(sample_means, bins=30, kde=True, ax=ax[1], color="seagreen")
    ax[1].axvline(sample_means.mean(), color="red", linestyle="--", label="Mean of means")
    ax[1].set_title(f"Sampling Distribution of Mean (n={n}) - approx. Normal")
    ax[1].legend()
    plt.tight_layout()
    plt.show()


exp_7()
```

Output

```
Population size : 10000
Population mean : 506.43
Population std  : 505.43

Sampling method comparison (sample mean of Spend)
Simple random (without replacement)    size=100  mean=  451.09
Simple random (with replacement)       size=100  mean=  428.91
Systematic                             size=100  mean=  545.59
Stratified by City (1% each)           size=100  mean=  451.47

City proportions - population vs stratified sample
         Population  Stratified
City
Chennai       0.403         0.4
Mumbai        0.299         0.3
Delhi         0.196         0.2
Pune          0.102         0.1

Sampling distribution of the mean (1000 samples, n=100)
Mean of sample means : 507.98  (population mean = 506.43)
Std of sample means  : 50.49
Theoretical SE = sigma/sqrt(n) : 50.54
```

Graph

